



# **DataBase HW 1**

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# 1. Answer Questions

- a. What the difference between type “char” and type “varchar”?

對於 char 來說，有沒有 var 的差別就在於有沒有定長，沒有 var 的 char 有固定長度，也就是輸入的英文長度未達固定長度時，會用空格來填充，有 var 的 varchar 再輸入長度未達時，並不會填充。因此 varchar 相較於 char 更省空間。

- b. Type “boolean” would be stored as which type in MySQL?

MySQL 中並沒有 boolean 這個型態，boolean 在裡面是作為 tinyint 的同義詞，0 視為 false，非 0 視為 true。

- c. How many bytes it should take for “tinyint”, “smallint”, “mediumint”, “int”? (e.g. 8 bytes for “bigint”) And what’s the range they can express? (e.g. from -1000 to 1000)

Tinyint: 1 byte, range: -128 ~ 127

Smallint: 2 byte, range: -32,768 ~ 32,767

Mediumint: 3 byte, range: -8,388,608 ~ 8,388,607

Int: 4 byte, range: -2,147,483,648 ~ 2,147,483,647

## 2. Tasks

### A. Create Tables

```
mysql> describe customers;
```

| Field                    | Type        | Null | Key | Default | Extra |
|--------------------------|-------------|------|-----|---------|-------|
| customer_id              | varchar(50) | NO   | PRI | NULL    |       |
| customer_unique_id       | varchar(50) | YES  |     | NULL    |       |
| customer_zip_code_prefix | int         | YES  |     | 0       |       |
| customer_city            | varchar(50) | YES  |     | NULL    |       |
| customer_state           | varchar(50) | YES  |     | NULL    |       |

```
5 rows in set (0.04 sec)
```

```
mysql> describe geolocation;
```

| Field                       | Type        | Null | Key | Default | Extra |
|-----------------------------|-------------|------|-----|---------|-------|
| geolocation_zip_code_prefix | varchar(50) | YES  |     | NULL    |       |
| geolocation_lat             | float       | NO   | PRI | NULL    |       |
| geolocation_lng             | float       | NO   | PRI | NULL    |       |
| geolocation_city            | varchar(50) | YES  |     | NULL    |       |
| geolocation_state           | varchar(50) | YES  |     | NULL    |       |

```
5 rows in set (0.00 sec)
```

```
mysql> describe order_items;
```

| Field               | Type        | Null | Key | Default | Extra |
|---------------------|-------------|------|-----|---------|-------|
| order_id            | varchar(50) | NO   | PRI | NULL    |       |
| order_item_id       | int         | NO   | PRI | 0       |       |
| product_id          | varchar(50) | YES  |     | NULL    |       |
| seller_id           | varchar(50) | YES  |     | NULL    |       |
| shipping_limit_date | datetime    | YES  |     | NULL    |       |
| price               | float       | YES  |     | NULL    |       |
| freight_value       | float       | YES  |     | NULL    |       |

```
7 rows in set (0.01 sec)
```

```
mysql> describe orders;
```

| Field                         | Type        | Null | Key | Default | Extra |
|-------------------------------|-------------|------|-----|---------|-------|
| order_id                      | varchar(50) | NO   | PRI | NULL    |       |
| customer_id                   | varchar(50) | YES  |     | NULL    |       |
| order_status                  | varchar(50) | YES  |     | NULL    |       |
| order_purchase_timestamp      | datetime    | YES  |     | NULL    |       |
| order_approved_at             | datetime    | YES  |     | NULL    |       |
| order_delivered_carrier_date  | datetime    | YES  |     | NULL    |       |
| order_delivered_customer_date | datetime    | YES  |     | NULL    |       |
| order_estimated_delivery_date | datetime    | YES  |     | NULL    |       |

```
8 rows in set (0.00 sec)
```

```
mysql> describe payments;
```

| Field                | Type        | Null | Key | Default | Extra |
|----------------------|-------------|------|-----|---------|-------|
| order_id             | varchar(50) | NO   | PRI | NULL    |       |
| payment_sequential   | int         | YES  |     | 0       |       |
| payment_type         | varchar(50) | YES  |     | NULL    |       |
| payment_installments | int         | YES  |     | 0       |       |
| payment_value        | float       | YES  |     | NULL    |       |

```
5 rows in set (0.01 sec)
```

```
mysql> describe products;
```

| Field                      | Type        | Null | Key | Default | Extra |
|----------------------------|-------------|------|-----|---------|-------|
| product_id                 | varchar(50) | NO   | PRI | NULL    |       |
| product_category           | varchar(50) | YES  |     | NULL    |       |
| product_name_length        | int         | YES  |     | 0       |       |
| product_description_length | int         | YES  |     | 0       |       |
| product_photos_qty         | int         | YES  |     | 0       |       |
| product_weight_g           | int         | YES  |     | 0       |       |
| product_length_cm          | int         | YES  |     | 0       |       |
| product_height_cm          | int         | YES  |     | 0       |       |
| product_width_cm           | int         | YES  |     | 0       |       |

```
9 rows in set (0.00 sec)
```

```
mysql> describe sellers;
+-----+-----+-----+-----+-----+-----+
| Field                | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| seller_id            | varchar(50)   | NO   | PRI | NULL    |       |
| seller_zip_code_prefix | int           | YES  |     | 0        |       |
| seller_city          | varchar(50)   | YES  |     | NULL    |       |
| seller_state         | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

## B. Query Tasks

1. Please list the number of all different clients. You must have **"COUNT"** syntax in usage of SQL.

```
mysql> source 1.sql
+-----+
| cnt  |
+-----+
| 96096 |
+-----+
1 row in set (0.32 sec)
```

2. Please list the number of different customer\_unique\_id which start with "ff9" . You must have **"DISTINCT"** syntax in usage of SQL.

```
mysql> source 2.sql
+-----+
| cnt  |
+-----+
| 35   |
+-----+
1 row in set (0.03 sec)
```

3. Please list which 3 cities have the most orders made in state 'RJ' . You must sort counts in decreasing order and have **"ORDER BY"** syntax in usage of SQL.

```
mysql> source 3.sql
+-----+
| city          |
+-----+
| rio de janeiro |
| niteroi       |
| nova iguacu   |
+-----+
3 rows in set (0.32 sec)
```

4. Please list the top 3 order\_id which cost the most days to deliver (the date of product arrival - the date the product starts to deliver). You must sort counts in decreasing order and have "ORDER BY" syntax in usage of SQL.

```
mysql> source 4.sql
+-----+
| order_id      |
+-----+
| 1b3190b2dfa9d789e1f14c05b647a14a |
| ca07593549f1816d26a572e06dc1eab6 |
| 285ab9426d6982034523a855f55a885e |
+-----+
3 rows in set (0.09 sec)
```

5. Please list the numbers of each states' orders which are not delivered.

```
mysql> source 5.sql
```

| state | cnt  |
|-------|------|
| SP    | 1245 |
| BA    | 124  |
| MG    | 281  |
| SC    | 91   |
| RJ    | 502  |
| RS    | 121  |
| CE    | 57   |
| MA    | 30   |
| PR    | 122  |
| PE    | 59   |
| DF    | 60   |
| PB    | 19   |
| PI    | 19   |
| MT    | 21   |
| GO    | 63   |
| SE    | 15   |
| ES    | 38   |
| PA    | 29   |
| MS    | 14   |
| AM    | 3    |
| AP    | 1    |
| RN    | 11   |
| TO    | 6    |
| RO    | 10   |
| AL    | 16   |
| RR    | 5    |
| AC    | 1    |

```
27 rows in set (0.04 sec)
```

6. There's a missing order in payment.csv, list its order\_id.

```
mysql> source 6.sql
```

| order_id                        |
|---------------------------------|
| bfb0f9bdef84302105ad712db648a6c |

```
1 row in set (0.79 sec)
```

7. Please list the order\_id which cost the most days to deliver (the date of product arrival - the date the product starts to deliver ) in each state. You must sort states in decreasing order and have **"ORDER BY"** syntax in usage of SQL.

```
mysql> source 7.sql
```

| customer_state | order_id                          |
|----------------|-----------------------------------|
| TO             | f5691c2b1ca263490374d13d020bd950  |
| SP             | 2aa91108853cecb43c84a5dc5b277475  |
| SE             | 285ab9426d6982034523a855f55a885e  |
| SC             | a622cb9542d08e3004c1a9a33a7cd859  |
| RS             | dfe5f68118c2576143240b8d78e5940a  |
| RR             | 2fa29503f2ebd9f53deba187160f3202  |
| RO             | 8489131c7dcc42a6d6d7df1be6802643  |
| RN             | 3566eabb132f8d64741ae7b921bbd10e  |
| RJ             | 1b3190b2dfa9d789e1f14c05b647a14a  |
| PR             | d1f789756b59a9fb40b3becadeac90e4  |
| PI             | 2fb597c2f772eca01b1f5c561bf6cc7b  |
| PE             | 031e7d4e559a1bf08e71a419aa998d0a  |
| PB             | 633f4fc75774571fccaeef584dfce5a7b |
| PA             | 440d0d17af552815d15a9e41abe49359  |
| MT             | 87f795f6abd4bbe786854091a9c6e651  |
| MS             | fc3830b4f000e216dc581df9a0e873e2  |
| MG             | c27815f7e3dd0b926b58552628481575  |
| MA             | 4fbc8d6f2f4db3e789d5a876fa349b56  |
| GO             | 2ba1366baecad3c3536f27546d129017  |
| ES             | ca07593549f1816d26a572e06dc1eab6  |
| DF             | 27167c20a18a136683606cb93c0168a9  |
| CE             | df6d8b7768a047c2981bae0a24afbb01  |
| BA             | 525e11b26fdb7f41471d289897d0f6da  |
| AP             | 437222e3fd1b07396f1d9ba8c15fba59  |
| AM             | 28f12a3e5117f5e70bbdb45be9188639  |
| AL             | d573a47c29466bb2e4e46aef0ff04f23  |
| AC             | 0b7cb1c7ea62d95ba39702f4ae850108  |

8. Please list the number of orders paid by debit card and purchase in 2017 in state 'RJ' .

```
mysql> source 8.sql
+-----+
| cnt |
+-----+
| 57 |
+-----+
1 row in set (0.03 sec)
```

9. Please list the top three seller\_ids who sold the most items priced over 150 and the number of these items they sold (price over 150). Only consider sellers who sold more than 100 items. (Hint: Use **WITH**)

```
mysql> source 9.sql
+-----+-----+-----+
| seller_id | cnt |
+-----+-----+-----+
| 4869f7a5dfa277a7dca6462dcf3b52b2 | 780 |
| 7c67e1448b00f6e969d365cea6b010ab | 487 |
| fa1c13f2614d7b5c4749cbc52fecda94 | 476 |
+-----+-----+-----+
3 rows in set (0.22 sec)
```

10. Please list the payment\_type that appears more than 100 times and the number they appear, **excluding the top two** most frequent payment methods. Order by the number the payment\_type appears.

```
mysql> source 10.sql
+-----+-----+
| payment_type | cnt |
+-----+-----+
| voucher | 2739 |
| debit_card | 1527 |
+-----+-----+
2 rows in set (0.06 sec)
```